

Sahithi Reddy Senagapally

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Education

Indiana University Bloomington Master of Science in Computer Science – GPA: 3.8/4.0 Relevant Coursework: Applied Algorithms, Computer Networks, Software Engineering, Cloud Computing TA: C Programming and Unix Gokaraju Rangaraju Institute of Engineering and Technology Bachelor of Technology in Computer Science and Engineering – GPA: 3.7/4.0 Relevant Coursework: Data Structures, Operating Systems, Database Management Systems, Artificial Intelligence, Machine Learning	Aug 2024 – May 2026 Aug 2019 – May 2023
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Technical Skills

Programming Languages: Java, Python, JavaScript, TypeScript, SQL, C, C++, C#, Bash, HTML, CSS Frameworks/Libraries: Spring Boot, Node.js, React, Angular, Next.js, Apache Kafka, FastAPI, Playwright Architecture: RESTful APIs, Microservices, Distributed Systems, Event-Driven Architecture, Object-Oriented Design, System Design Databases: PostgreSQL, MySQL, Oracle Database, DynamoDB, SQL Server, MongoDB, Redis Cloud/DevOps: AWS (Lambda, API Gateway, S3, IAM, EC2), Azure, GCP, Docker, Kubernetes, Terraform, GitHub Actions, CI/CD, Git Testing/Monitoring: JUnit, Mockito, Testcontainers, PyTest, k6, Grafana AI/ML: LangChain, RAG, OpenAI API, Anthropic API, PyTorch, NumPy, Claude Code, Codex
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Professional Experience

Indiana University Bloomington Software Engineer • Redesigned enrollment processing for a course-management platform serving 10,000+ users to handle concurrent requests safely, introducing transactional MySQL capacity checks and concurrency controls that cut duplicate enrollment conflicts by ~99% and prevented inconsistent course-capacity updates. • Delivered 8+ Node.js REST APIs for course and user-management workflows, adding role-based authorization and cutting high-traffic endpoint response time by 25% through SQL query tuning, index optimization, and server-side pagination.	Oct 2024 – May 2026 Bloomington, IN
Isomerous Software Developer Intern • Designed and deployed a serverless AWS backend using API Gateway, Python Lambda, DynamoDB, S3, and IAM, replacing a manual product-inspection workflow with an API-driven ingestion pipeline and reducing processing turnaround time by 87% . • Developed Lambda functions for payload validation, metadata transformation, image processing, and coordinated S3/DynamoDB persistence, enabling traceable retrieval and scaling the ingestion pipeline to 20,000+ submissions/day.	Jun 2025 – Aug 2025 Ashburn, VA
Tata Consultancy Services Software Engineer • Engineered Java/Spring Boot rule-engine services for AML transaction monitoring across 1M+ financial transactions daily, improving fraud-detection true positives by 83% through configurable risk-pattern evaluation and optimized rule execution. • Cut real-time alert latency by 75% , from 12 to 3 minutes , addressing a critical requirement from 4 banking clients by optimizing Oracle SQL, adding indexes, and resolving multithreading bottlenecks; improvements contributed to onboarding 2 additional clients. • Built a Kafka-based event-driven monitoring service with consumer groups, retry handling, idempotent consumers, and ordering safeguards, replacing batch processing with a fault-tolerant streaming pipeline; validated with JUnit and Spring Boot integration tests. • Automated regulatory watchlist ingestion and reconciliation using validation, deduplication, and optimized Oracle database operations, boosting daily processing throughput by 40% .	May 2023 – Jul 2024 Bangalore, India
ADP Software Engineer • Architected backend reporting APIs that aggregated employee data across internal services, implementing search, sorting, pagination, and response transformations to minimize payload size and support responsive on-demand reporting. • Improved Report Builder load time by 77% , from 8.4s to 1.9s , for 1,000+ users by consolidating redundant API requests, parallelizing downstream calls, and streamlining React rendering and integrations with authentication and payments services.	July 2022 – Nov 2022 Hyderabad, India

Projects

Distributed Payment Processing & Ledger Platform - GitHub • Built a distributed payment and double-entry ledger platform in Java/Spring Boot with Kafka-based async workflows, PostgreSQL, and Redis, sustaining 3,600+ transactions/min at 12.6ms p95 latency with 0% failed requests under k6 load testing. • Implemented a transactional outbox pattern with idempotent consumers and consistent lock ordering to prevent duplicate processing and eliminate deadlocks; validated with Testcontainers and Mockito tests, deployed via GitHub Actions CI/CD.	Sept 2026
AI Agent for Legacy System Automation - GitHub • Developed an AI agent that operates legacy, API-less bank/credit-union back-office UIs by controlling a real browser via Playwright, using LLM-driven discovery to convert successful task runs into reusable, typed "capability artifacts." • Designed a deterministic replay engine that executes learned capability artifacts in production without re-invoking the LLM, backed by Pydantic schema validation, an error taxonomy, human-escalation pathways, and structured evidence logging.	Aug 2026